Image Classification of Forest Fires Using Machine and Deep Learning

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***Abstract*—With the increasing threat presented by climate change, forest fires are becoming more destructive and frequent. These catastrophes present immense danger to our forest ecosystems, infrastructure, and overall health. To dispatch resources in a timely manner, quick and accurate identification of these fires is crucial. Here, machine and deep learning can be leveraged as a solution. In this research, we investigate the classification abilities of Support Vector Machines (SVM), Xception, ResNet50, and MobileViT. In addition, two transfer learning approaches making use of Xception and ResNet50 as the backbone networks are explored. The popular DeepFire dataset is utilized for the training and comparison of model performance. After tuning each architecture, our results indicate the transfer learning approach with Xception as the backbone provides the highest accuracy obtaining 99.211%.**

***Index Terms*—Deep learning, Forest fire, Transfer learning, SVM, ResNet, Xception, MobileViT**

I. Introduction

Forest fires are destructive natural phenomena that are often characterized by uncontrollable and unpredictable fires fueled by large amounts of vegetation. These fires have been increasing in both acres burned and frequency according to yearly wildfire statistics provided by the National Centers for Environmental Information. The record for most acres burned is attributed to the year 2020 with an amount of 10 million acres in the United States [1]. With the increasing threat of global climate change, these fires have the potential to become increasingly frequent and disruptive. In a study of climate and forest fires, Flannigan, Stocks, and Wotton investigated the impact of climate change on fire severity. Using the proposed estimation models, they show a prediction of up to 50% increase in seasonal severity rating further feeding these disasters [2]. In addition to the ecological damage, these fires produce vast amounts of particulate which is smaller than 2.5 micrometers leading to increased risk for health complications [3]. Furthermore, the aftermath of these fires creates burn scars inducing an influx of debris to our natural water sources impacting overall water quality [4]. For these reasons, early and accurate detection of these disasters is paramount to reduce impact.

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One area of interest in solving this complex problem is machine learning. Machine learning at the core represents a shift in engineering perspective where model and algorithm development are substituted for large datasets of example classes. Here, the model can then generate a function or estimate which is valuable in many applications [5]. Deep learning takes machine learning a step further utilizing neural networks and increasingly large layer counts to produce function approximations. Additionally, modifications to these approaches have yielded many popular architectures for visual inspection including Convolutional Neural Networks (CNN) [6]. In this paper, the architectures investigated for fire detection include Support Vector Machine (SVM) and three CNN architectures. The following section investigates existing work in the area of fire detection. Then, we discuss the approaches of our research in detail. After which, the simulations conducted are introduced and the results are discussed. Finally, conclusions are drawn and future challenges are presented and addressed.

II. Literature Review

With the increasing amount of volatile forest fires, a multitude of approaches have been investigated. This includes wireless sensor networks, remote sensing, and visual inspection methods [7]. With the recent advancements in computational performance for deep learning, computer vision inspection approaches are gaining popularity. In this area, classification methods have been developed using popular architectures. One such approach used VGG-19 and outperformed K-nearest neighbors, Support Vector Machine (SVM), and logistic regression by achieving 95% accuracy [8]. Expanding on this work, a new architecture was investigated with transfer learning creating FFireNet which increased the reported accuracy to 98.42% [9]. A similar approach investigated Residual Networks and VGG with varying layer counts providing an accuracy of 99.50% on the utilized dataset [10].

While classification is an important aspect of visual inspection, novel architectures in object detection are also being applied in this area. One approach made use of the You Only Look Once (YOLO) version 4 architecture and enhanced the backbone with MobileNet [11]. In a similar approach, a YOLOv3 architecture was modified using EfficientNet and depth-wise convolutions to achieve increased accuracy in small fire detection [12]. In this case, the performance metrics of importance are size and inference speed to achieve close to real-time performance in identification systems. Our research focuses on the classification of forest fires and is a detailed analysis of what was reported in our recent work on desert and forest fire problems [13]. By leveraging the DeepFire dataset, we seek to further improve upon detection accuracy of forest fires through the proposed transfer learning approach.

III. Architecture Overview

A. Support Vector Machine

SVM is a concept that was introduced in 1979 and was originally for the task of binary classification. This approach maps the training data into higher dimensions with a kernel and attempts to minimize the margin between classes to produce a decision boundary [14]. As the only non-deep learning approach, this model serves as a baseline for comparison against the larger and more advanced architectures in our research.

B. ResNet50

Residual Networks were created by He et al. in 2015 with the challenges of deeper networks at the forefront of their research [15]. With the increasing layer count resulting in backpropagation loss, skip connections were proposed as a solution. This allows a path for the gradients to reach earlier layers in the architecture vital for training deeper and larger models. The residual family of architectures consists of many different layer counts including 34, 50, 101, and 152 variants. In our research, the 50-layer model was chosen as it provides sufficient performance on smaller datasets.

C. Xception

Xception is a CNN architecture that leverages depthwise separable convolutions coupled with a pointwise convolutional filter for a large reduction in computational costs [16]. Additionally, this model is 71 layers deep and leverages skip connections which have been shown to outperform the earlier inception models on large-scale datasets [16]. These skip connections also help alleviate the vanishing gradient problem present in deeper models by creating a pathway for loss information to reach earlier layers.

D. MobileViT

Transformers have been heavily used in the area of natural language processing, however, in recent years these have seen rapid growth in computer vision applications [17]. Through the combination of these vision transformers (ViT) and convolutions, MobileViT was created. This architecture combines the global long-range dependencies and relationships of the transformer with the spatial representation of convolution into a small mobile-ready architecture [18]. Due to the smaller size and emphasis on mobile deployment, MobileViT will provide a unique comparison in terms of model size.

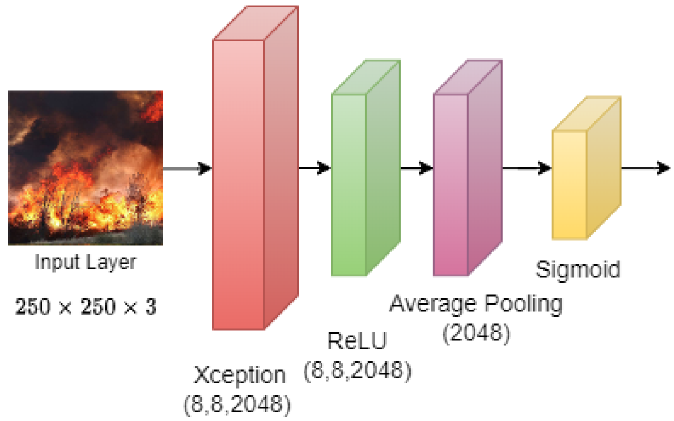

Fig. 1. TL-Xception architecture.

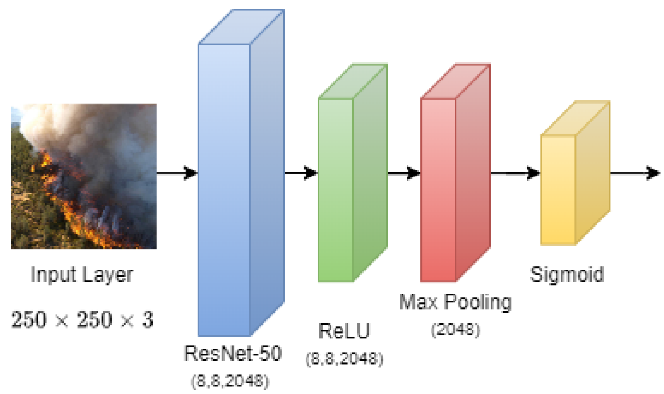

Fig. 2. TL-ResNet50 architecture.

E. Transfer Learning (TL)

Transfer learning is the approach of applying previously learned weights on large-scale datasets to new problems. This allows the early layers utilized for feature extraction to be frozen and determined by large datasets. Then, through additional layers and fine-tuning new data, the application-specific model can be created [19]. In our research, this is applied to the Xception and ResNet50 architectures to formalize two new modified architectures by making use of the ImageNet dataset weights. The base network now loaded with new weights is then frozen to prevent these weights from changing. New predictive layers are added in the form of Rectified Linear Unit (ReLU), Pooling, and a final dense layer with sigmoid activation. Fig. 1 and Fig. 2 illustrate the modified ResNet50 and Modified Xception architectures with the additional predictive layers.

IV. Simulations and Results

The following section discusses the simulations and experiments conducted to train, tune, and compare model performance. These simulations were conducted using the following software libraries and hardware: TensorFlow 2.10, Keras 2.10,

TABLE I
DEEPFIRE DATASET SPLIT [8].

Label	Train	Validation	Test	Total
Fire	608	152	190	950
No-Fire	608	152	190	950
Total	1216	304	380	1900

TABLE II
THE AUGMENTATION APPLIED TO TRANSFER LEARNING MODELS.

Random Augmentation Type	Range
Rotation	0°-50°
zoom	± .1%
Shear	.1%
Translation	.1% - .2%

and Python 3.9 on an Intel® Core i5-7600K, 16 GB DDR4, NVIDIA GeForce GTX 1070 with 8GB GDDR5.

A. Dataset and Augmentation

To train and compare model performance, large amounts of data are required. For this reason, the popular public dataset DeepFire has been leveraged [8]. This dataset contains 1900 evenly split images between fire and no-fire of size 250 $\times$ 250. With the primary focus of this dataset on outdoor forest environments, it was chosen to gauge the performance of the proposed models. Tab. I shows the symmetrical split used for the simulations in this paper along with the corresponding values for each class, fire or no-fire. Additionally, a sample of this dataset can be visualized in Fig. 3 with the corresponding class label.

Data augmentation was used in the simulations for the transfer learning models to better generalize the fire features and avoid over-fitting, where the model learns the training data too well and cannot infer on unseen data. Caused by the limited trainable parameters, these models would quickly over-fit to the training data without this pre-processing step. This was accomplished through the Keras library which allows parameters to be set to alter the images as they are fed to the model. This method increases the variation of images that the model trains with which fights over-fitting and leads to a more robust model. In our simulations, the following augmentation was utilized, rotation, zoom, shear, and translation. The selected values for each setting are shown in Tab. II These parameters were kept consistent throughout each transfer learning model training and tuning simulations, however, the augmentation is not active during the testing process.

B. Hyperparameter Search

Each model presented in this research utilizes different parameters and settings to tune the performance. To obtain an optimal state for each model, hyper-parameter tuning was conducted to search and compare these values. This section discusses the search space used and the resulting parameters found for the final comparison of the dataset.

Fig. 3. Sample images from the DeepFire dataset [8].

TABLE III
SVM RESULTS.

C Value	Kernel	Test Accuracy
100	**Poly-Degree 2**	**93.684%**
1	RBF	93.421%
10	Poly-Degree 2	92.895%
0.1	Linear	92.368%

1) SVM: To implement SVM for fire detection and classification, the Scikit-Learn library was leveraged [20]. The search space for this algorithm consisted of the C value, kernel function, and degree of polynomial for polynomial kernels. The remaining values were kept at their default. The C value determines the size of the corresponding margin of this classifier, a smaller value achieves a larger margin and a large value creates smaller margins. The kernel is a function that is applied to the input values to transform them into higher dimensionality leading to an easier separation. The values investigated in our search consist of Linear, Radial Basis Function (RBF), Polynomial, and Sigmoid. Additionally, the polynomial kernel was searched with degree values of 1, 2, and 3.

With the above parameters selected, a grid search was conducted to compare performance. Tab. III shows the best results on the test data for each corresponding C value. Here, the larger C value of 100 provided the best performance with an accuracy of 93.684%. Additionally, this was achieved with the polynomial kernel of degree 2.

2) ResNet50: To determine optimal performing parameters for the deep learning architectures, KerasTuner was used which provides a systematic method for training and comparing different specified settings [21]. Here, five optimization algorithms were tested and compared including Adam, Nadam, Adagrad, RMSprop, and SGD. Each of these optimization algorithms has unique parameters to assist in the convergence of the models weights and were also included in the search. This consisted of learning rate, momentum, β_1, β_2, and ρ. The corresponding highest accuracy parameters were then selected, trained, and tested to compare performance.

These corresponding top 3 optimization algorithms and parameters found for ResNet50 through this search are outlined in Tab. IV. Here, it can be seen the SGD and RMSprop optimization algorithms provided the highest accuracy for this model achieving 96.842% on the test data. This was followed closely by the Nadam optimizer which obtained 96.579% overall. Following these simulations, dropout was also investigated to provide better generalization. Here, the search conducted selected random percentages to drop during training. The results in this simulation indicated 5% provided the best accuracy however, for each run in Tab. IV the accuracy was unchanged from no dropout.

3) Xception: The resulting performance of the Xception architecture is shown in Tab. V. The optimization algorithms found through the simulation included Adam, SGD, and Nadam. These three algorithms were close in accuracy all achieving above 96% on the test data. The best performance was found to be Adam with 97.632% accuracy. Additional performance gains were found on the Xception architecture with the dropout search simulations. A value of 40% was found to provide the largest gains in terms of accuracy which achieved 98.421% overall with the Adam optimization settings. This is a gain of 0.789% accuracy over the non-dropout run which is a consistent trend with each optimizer in our search for Xception.

4) MobileViT: Following the same tuning methodology, the results found for the MobileViT architecture are shown in Tab. VI. Here, the simulations to determine the best dropout were not conducted due to the unique architecture combining transformers and convolutions. The best performance with this architecture was found to be Adagrad with a sizeable learning rate of 0.1 which obtained 96.570% accuracy. This was followed by SGD and Nadam which reached 95% and 94.474% accuracy, respectively.

5) Transfer learning Models: Following the simulations on Xception and ResNet50, transfer learning models were created utilizing these architectures as the feature extraction layer. The imagenet weights, a massive image dataset with 1000 classes, were loaded into each model and then frozen to prevent changes during fine-tuning. Then additional predictive layers were added to the architectures including a ReLU non-linearity, max/average pooling, and a single neuron dense layer with sigmoid activation for binary classification. The hyperparameters chosen for these models were taken directly from the base model search results of Xception and ResNet50, respectively. These values along with the specified epochs, batch size, and learning rate can be seen in Tab. VII.

The performance benefit of transfer learning is realized in Tab. VIII. Here, both the modified Xception and modified ResNet50 models achieved higher accuracy than their nontransfer learning variants. The modified Xception model achieved the highest accuracy with 99.211% overall on the test set followed closely by modified ResNet50 obtaining 98.684%. In the pursuit of increased accuracy for precise fire detection, transfer learning, and fine-tuning can bring favorable performance gains.

Fig. 4. TL-ResNet50 DeepFire performance.

C. Results

Upon successful tuning of each architecture in our research, each model was trained with the specified parameters and settings. Tab. VIII draws a full comparison of the architectures on the test set. Here, the modified Xception architecture making use of transfer learning achieved the highest accuracy with 99.211%. The other deep learning architectures in our research each obtained notable accuracies over 96%, as well. Additionally, the SVM approach, which utilizes separation boundaries and higher dimensionality mappings, still achieved notable accuracy at 93.684%.

A confusion matrix is constructed for each of the modified transfer learning architectures to further visualize their classification performance. Fig. 4 maps the classification of the modified ResNet50 model on the test data. Here, it can be seen that the model only misclassified 5 images, specifically 5 images were labeled no fire and the model predicted fire. In comparison, the modified Xception model achieved only 3 false classifications as seen in Fig. 5. It is shown that this model correctly classified all no-fire images and falsely labeled only 3 fire images overall. Our results show that while traditional machine learning algorithms provide good accuracy, deep learning can leverage deeper feature extraction and achieve higher accuracy. This can be further extrapolated by utilizing larger dataset weights in a transfer and tune approach which helps achieve faster convergence and accuracy.

Finally, the modified transfer learning models are compared against recently reported results in the literature on the DeepFire dataset shown in Tab. IX. Here, the modified Xception model achieves the highest accuracy with 99.211%. This comparison further illustrates the power of transfer learning and its application in obtaining higher accuracy through the distillation of existing knowledge.

TABLE IV
RESNET50 TOP THREE OPTIMIZATION ALGORITHMS WITH THE BEST-PERFORMING HYPERPARAMETERS.

Dropout	Optimizer	Learning Rate	Momentum	ρ	β_1	β_2	Test Accuracy
5%	SGD	0.01	0.03	-	-	-	**96.842%**
0%	SGD	0.01	0.03	-	-	-	**96.842%**
5%	RMSprop	0.0001	0	0.89	-	-	**96.842%**
0%	RMSprop	0.0001	0	0.89	-	-	**96.842%**
5%	Nadam	0.001	-	-	0.9	0.9	96.579%
0%	Nadam	0.001	-	-	0.9	0.9	96.579%

TABLE V
XCEPTION TOP THREE OPTIMIZATION ALGORITHMS WITH THE BEST-PERFORMING HYPERPARAMETERS.

Dropout	Optimizer	Learning Rate	Momentum	β_1	β_2	Test Accuracy
40%	**Adam**	**0.001**	**-**	**0.9**	**0.999**	**98.421%**
40%	SGD	0.01	0.8	-	-	97.895%
40%	Nadam	0.001	-	0.9	0.999	97.884%
0%	Adam	0.001	-	0.9	0.999	97.632%
0%	SGD	0.01	0.8	-	-	97.368%
0%	Nadam	0.001	-	0.9	0.999	96.842%

TABLE VI
MOBILEVIT TOP 3 OPTIMIZATION ALGORITHMS WITH BEST-PERFORMING HYPERPARAMETERS.

Optimizer	Learning Rate	Momentum	β_1	β_2	Test Accuracy
Adagrad	**0.1**	**-**	**-**	**-**	**96.579%**
SGD	0.01	0.1	-	-	95.000%
Nadam	$1e-5$	-	0	0	94.474%

TABLE VII
TRANSFER LEARNING TRAINING PARAMETERS.

Setting	TL-ResNet	TL-Xception
Batch Size	64	64
Epochs	50	50
Optimizer	RMSprop	Adam
Learn Rate	0.001	0.001

Fig. 5. TL-Xception DeepFire performance.

TABLE VIII
BEST PERFORMANCE ON DEEPFIRE TEST DATA.

Architecture	Accuracy (%)	Precision	Recall	F1-score
TL-Xception	**99.211**	**1.0**	**0.9842**	**0.9920**
TL-ResNet50	98.429	0.979	0.989	0.989
Xception	98.421	0.9742	0.9947	0.9844
ResNet50	96.842	0.9684	0.9684	0.9684
MobileViT	96.579	0.9538	0.9789	0.9662
SVM	93.684	0.937	0.937	0.937

V. FURTHER DISCUSSION

One noticeable obstacle of the TL approach is the reduction in the trainable parameters available within the network. In the proposed TL-Xception architecture, the trainable parameters exist solely in the predictive layers added. This reduces the trainable parameters from ≈ 20 million in the base Xception to 2049 in the proposed TL-Xception. Because of this drastic decrease in trainable weights, the TL networks struggled to generalize due to over fitting. To account for this, image augmentation was explored to increase image variety to the model allowing a greater generalization of fire features.

VI. CONCLUSION

With the increasing temperatures and abnormal weather events driven by climate change, forest fires are an inevitable side effect. These catastrophes are increasing in frequency, size, and volatility. To identify and dispatch resources efficiently, it is crucial to mitigate these disasters before they become uncontrollable. To assist in this effort, the investigation of machine and deep learning methods for forest fire classification was conducted. Through hyperparameter tuning, an optimal performing setting was achieved for each of the models in our research. Furthermore, transfer learning was investigated for the Xception and ResNet50 backbone networks. Our results show that the transfer learning models achieved higher accuracy over their base networks with a modified Xception model obtaining a very high 99.211% accuracy. This

TABLE IX
COMPARISON OF RECENTLY PUBLISHED RESULTS ON DEEPFIRE.

Method	Accuracy	Precision	Recall	F1-Score
TL-Xception	**99.211%**	**1.000**	**0.9842**	**0.9920**
Idroes et al., 2023 [22]	98.680 %	0.9947	0.9793	0.9868
Khan & Khan, 2022 [9]	98.421%	0.9742	0.9947	0.9844
Modified ResNet50	98.421%	0.9792	0.9895	0.9844
Mousavi & Ilanloo, 2023 [23]	96.880%	0.9736	0.969	0.9861
Khan et al. 2022 [8]	95.000%	0.9572	0.9421	0.9496
Supriya & Gadekallu, 2023 [24]	94.470%	n/a*	n/a*	n/a*

* metrics were not available

model also outperformed recent publications on the DeepFire dataset showing a promising trend in the classification of forest fires.

VII. FUTURE WORK

While the classification of forest fire plays an important role in solving for these disasters, there still remain challenges that will need to be addressed. This includes simplifying the models utilized to approach real-time performance. Because fires are quick to spread, a real-time approach is vital in addressing the needs of this area. Additionally, with the nature of these remote environments, data transmission is also an area of challenge. The authors plan on addressing the real-time needs through emerging object detection methods.

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